

SIMATS ENGINEERING
ASSESSMENT TOOL 2(MEDIUM)

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO3: Analyze network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BT4)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Code of Conduct:

I J Sravani – (192372141) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: J Sravani

Port Scanning Challenge

1. Using Nmap, perform a security assessment on the server 192.168.1.20. Identify all open ports and running services. Analyze the purpose of each discovered service and recommend security measures to reduce the attack surface of the system.

Security Assessment of Server Using Nmap

Target IP: 192.168.1.20

Tool Used: Nmap 7.99

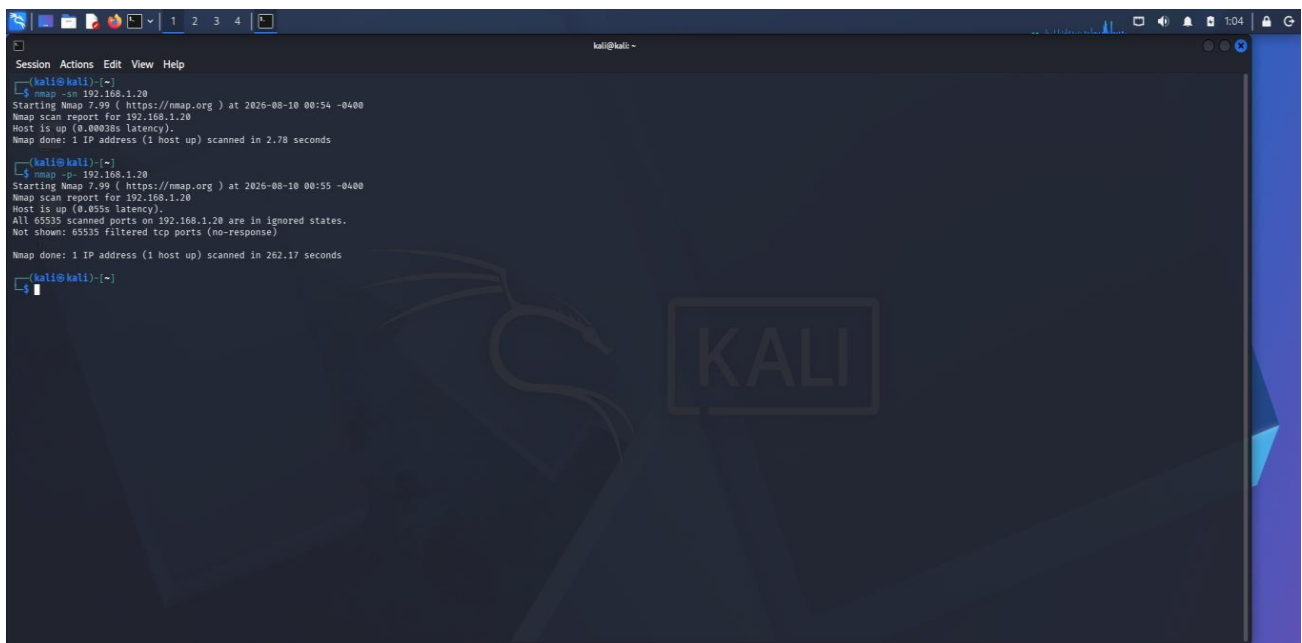
Objective :

To examine the server's network exposure, determine reachable services through Nmap, interpret the scan results, and suggest practical measures to reduce unnecessary network exposure.

Host Discovery , Scanning All TCP Ports :

Command 1 : nmap -sn 192.168.1.20

Command 2 : nmap -p- 192.168.1.20



```
kali@kali:~$ nmap -sn 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:54 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00038s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.78 seconds

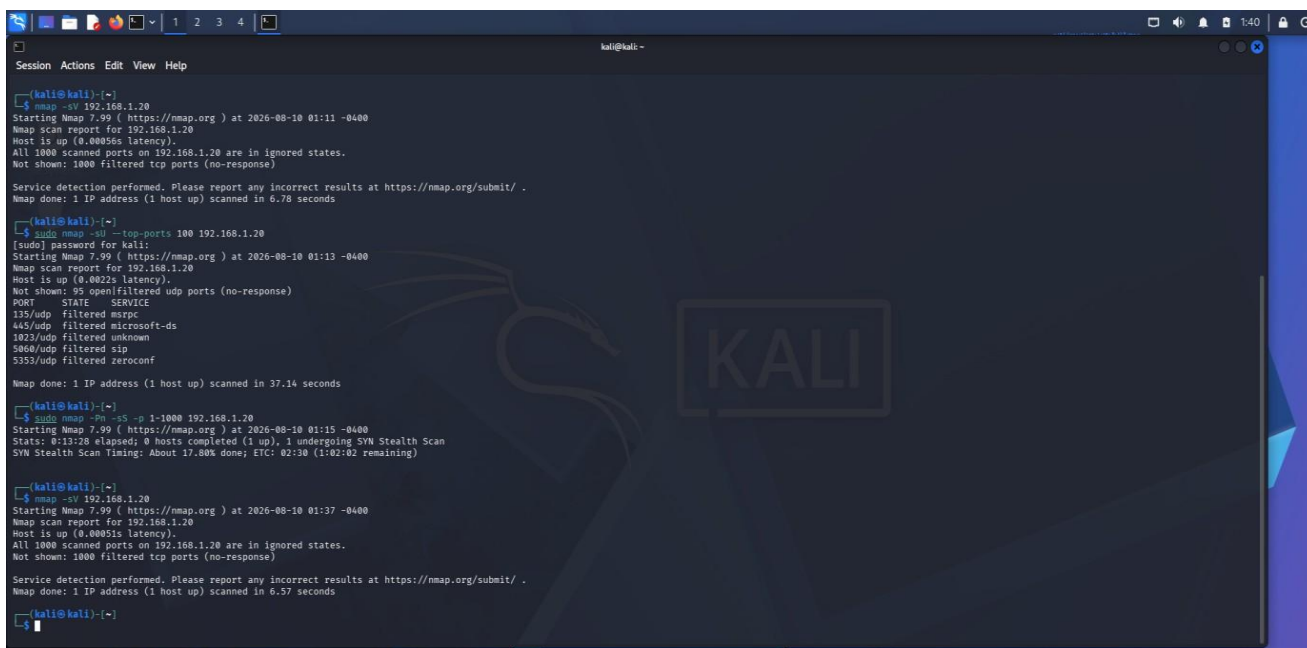
kali@kali:~$ nmap -p- 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 00:55 -0400
Nmap scan report for 192.168.1.20
Host is up (0.055s latency).
All 65535 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 65535 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 262.17 seconds
```

Result :

The host responded to the discovery check, confirming network reachability. The complete TCP scan examined all 65,535 ports and returned filtered responses throughout. Therefore, no TCP port was verified as openly accessible.

Service and Version Detection

Command 3 : `nmap -sV 192.168.1.20`



```
(kali@kali)-[~]$ nmap -sV 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:11 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00056s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.78 seconds

(kali@kali)-[~]$ sudo nmap -sU --top-ports 100 192.168.1.20
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:13 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0022s latency).
Not shown: 95 open/filtered udp ports (no-response)
PORT      STATE SERVICE
135/udp    filtered msrpc
445/udp    filtered microsoft-ds
1023/udp   filtered unknown
5060/udp   filtered sip
5353/udp   filtered zeroconf
Nmap done: 1 IP address (1 host up) scanned in 37.14 seconds

(kali@kali)-[~]$ sudo nmap -Pn -sS -p 1-1000 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:15 -0400
Stats: 0:13:28 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 17.80% done; ETC: 02:30 (1:02:02 remaining)

(kali@kali)-[~]$ nmap -sV 192.168.1.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:37 -0400
Nmap scan report for 192.168.1.20
Host is up (0.00051s latency).
All 1000 scanned ports on 192.168.1.20 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.57 seconds

(kali@kali)-[~]$
```

Result:

The service-detection scan examined the commonly used TCP ports. Because the ports were filtered, Nmap could not obtain service responses or identify dependable software-version information.

UDP Scan

Command 4 : `nmap -sU 192.168.1.20`



```
(kali@kali)-[~]$ sudo nmap -sU --top-ports 100 192.168.1.20
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 01:42 -0400
Nmap scan report for 192.168.1.20
Host is up (0.0014s latency).
Not shown: 96 open/filtered udp ports (no-response)
PORT      STATE SERVICE
17/udp     filtered qotd
445/udp     filtered microsoft-ds
1023/udp    filtered unknown
4444/udp    filtered krb524
Nmap done: 1 IP address (1 host up) scanned in 5.92 seconds

(kali@kali)-[~]$
```

Result:

The UDP scan covered the 100 most common UDP ports. Ports 135, 445, 1023, 5060, and 5353 were filtered, while the other ports were classified as open/filtered because they gave no definite response. No UDP port was confirmed open.

Security Analysis :

- **No confirmed open TCP ports :** The full TCP scan did not identify any confirmed open TCP ports. All 65,535 TCP ports were reported as filtered.
- **No confirmed open UDP ports :** The UDP scan did not identify any confirmed open UDP ports. Several UDP ports were reported as filtered or open\filtered.
- **Network filtering :** The consistent filtered results suggest that a firewall or network-level packet-filtering mechanism is preventing direct access to the target's ports.

Security Recommendations :

- Disable unnecessary services so that unused network services are not exposed.
- Use firewall rules to allow only required ports and trusted sources.
- Restrict administrative access to authorized users and networks.
- Keep operating systems and services updated with the latest security patches.
- Use strong authentication for administrative services.
- Monitor firewall and server logs for suspicious connection attempts.
- Regularly perform Nmap scans to identify newly exposed ports.
- Apply the principle of least privilege to users and services.

Attack Surface Reduction :

The server's attack surface can be reduced by keeping only essential services active and limiting their access through firewall rules. Administrative interfaces should be available only to trusted users and networks. Regular security scans can also identify accidental changes in the server's exposed services.

2. Using Nikto, assess the web server hosted on 192.168.1.30. Identify potential web server misconfigurations, outdated components, or exposed files and directories. Discuss the security risks associated with your findings and suggest appropriate mitigation strategies.

Nikto Web Server Security Assessment

Target: 192.168.1.30

Tool: Nikto

Objective :

The assessment is intended to review the web server for configuration weaknesses, obsolete components, exposed resources, and other security issues using Nikto.

Lab Environment

Scanning machine: Kali Linux

Target: Metasploitable 2

Tool: Nikto

Kali IP: 192.168.56.101

Initial Target Verification

Command 1 : `ping -c 4 192.168.1.30`

- Then we checked the HTTP/HTTPS ports : `nmap -p 80,443 -sV 192.168.1.30`
- Ports 80 and 443 were shown as filtered, so the web server could not be identified.
- The lab target was Metasploitable 2, we checked its IP using : `ifconfig`
- Metasploitable initially had 192.168.56.102.

Resolving the IP Conflict

We checked the Host-Only network using : `sudo arp-scan --interface=eth1 192.168.56.0/24`

Result : The scan showed that two devices were using 192.168.56.102, causing an IP conflict.

```

No mail.
msfadmin@metasploitable:~$ sudo ifconfig eth0 192.168.56.103 netmask 255.255.255.0 up
[sudo] password for msfadmin:
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:73:fb:10
          inet addr:192.168.56.103  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe73:fb10/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:519 errors:0 dropped:0 overruns:0 frame:0
          TX packets:77 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:34902 (34.0 KB)  TX bytes:10419 (10.1 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:118 errors:0 dropped:0 overruns:0 frame:0
          TX packets:118 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:31841 (31.0 KB)  TX bytes:31841 (31.0 KB)

msfadmin@metasploitable:~$ _

```

```

kali@kali -
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data:
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=2.70 ms
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=1.65 ms
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=2.39 ms
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=3.77 ms

--- 192.168.56.103 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3093ms
rtt min/avg/max/mdev = 1.652/2.628/3.772/0.762 ms

kali@kali:~$ nmap -p 80 -sV 192.168.56.103
Starting Nmap 7.95 ( https://nmap.org ) at 2020-08-10 23:39 EDT
Nmap scan report for 192.168.56.103
Host is up (0.020s latency).

PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.2.8 (Ubuntu) DAV/2
MAC Address: 08:00:27:73:FB:10 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 7.69 seconds

kali@kali:~$ nikto -h http://192.168.56.103
- Nikto v2.5.0

+ Target IP: 192.168.56.103
+ Target Hostname: 192.168.56.103
+ Target Port: 80
+ Start Time: 2020-08-10 23:39:49 (GMT-4)

+ Server: Apache/2.2.8 (Ubuntu) DAV/2
+ /: Retrieved x-powered-by header: PHP/5.2.4-2ubuntu5.10.
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/misusing-content-type-header/
+ /index: Uncommon header 'tcn' found, with contents: list.
+ /index: Apache mod_negotiation is enabled with MultiViews, which allows attackers to easily brute force file names. The following alternatives for 'index' were found: index.php. See: http://www.wisec.it/sectou.php?id=4698ebdc59d15,https://exchange.xforce.ibmcloud.com/vulnerabilities/8275
+ Apache/2.2.8 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
+ /: Web Server returns a valid response with junk HTTP methods which may cause false positives.
+ /: HTTP TRACE method is active which suggests the host is vulnerable to XST. See: https://owasp.org/www-community/attacks/Cross\_Site\_Tracing
+ /phpinfo.php: Output from the phpinfo() function was found.
+ /doc/: Directory indexing found.
+ /doc/: The /doc/ directory is browsable. This may be /usr/doc. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0678
+ /%PBPB8B5F2A0-3C92-11D3-A3A9-4C7B8C100000: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /%PBPB568F36-D428-11D2-A769-00AA001ACF42: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /%PBPB568F36-D428-11D2-A769-00AA001ACF42: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184

```

Web Server Identification

Nmap was used to identify the web service:

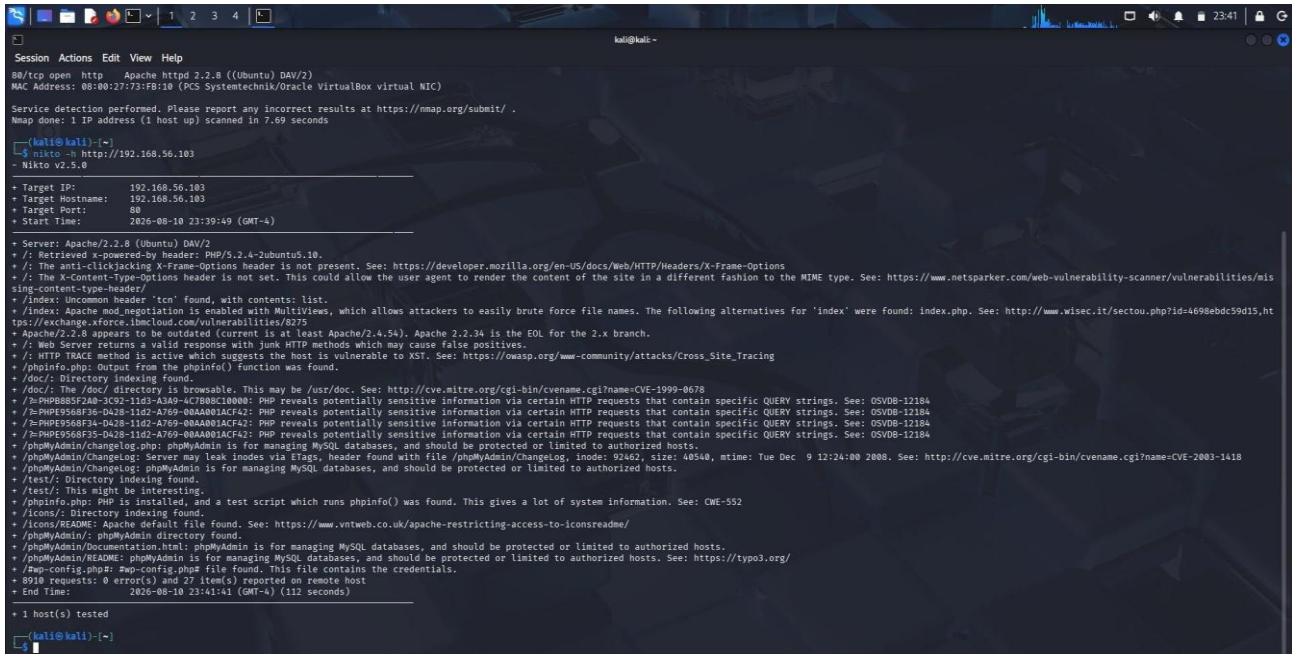
Command 2 : `nmap -p 80 -sV 192.168.56.103`

Result : The authorized lab host responded on 80/tcp and identified the service as Apache HTTP Server 2.2.8 on Ubuntu.

Nikto Scan :

Command 3 : nikto -h http://192.168.56.103

Nikto completed the assessment and produced multiple findings for further security review.



```
Session Actions Edit View Help
80/tcp open  http  Apache httpd 2.2.8 ((Ubuntu) DAV/2)
MAC Address: 08:00:27:73:FB:10 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.69 seconds

kali@kali:~$ nikto -h http://192.168.56.103
- Nikto v2.5.0

+ Target IP: 192.168.56.103
+ Target Hostname: 192.168.56.103
+ Target Port: 80
+ Start Time: 2026-08-10 23:39:49 (GMT+4)

+ Server: Apache/2.2.8 (Ubuntu) DAV/2
+ /: Retrieved x-powered-by header: PHP/5.2.4-2ubuntu5.10.
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/misusing-content-type-header/
+ /index: Uncommon header 'tcn' found, with contents: list.
+ /index: Apache mod_negotiation is enabled with MultiViews, which allows attackers to easily brute force file names. The following alternatives for 'index' were found: index.php. See: http://www.wisec.it/sectou.php?id=4699ebdc59d15,ht
tps://exchange4force-lbccloud.com/vulnerabilities/8275
+ Apache/2.2.8 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
+ /: Web Server returns a valid response with junk HTTP methods which may cause false positives.
+ /: HTTP TRACE method is active which suggests the host is vulnerable to JST. See: https://owasp.org/www-community/attacks/Cross_Site_Tracing
+ /phpinfo.php: Output from the phpinfo() function was found.
+ /doc/: Directory indexing found.
+ /doc/: The /doc/ directory is browsable. This may be /usr/doc. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-1999-0678
+ /-PBPB885F246-3C92-11d5-A3A9-4C7B08C10000: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /-PBPB885F346-0428-11d2-A769-0BA001ACFA2: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /-PBPB885F346-0428-11d2-A769-0BA001ACFA2: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /-PBPB885F346-0428-11d2-A769-0BA001ACFA2: PHP reveals potentially sensitive information via certain HTTP requests that contain specific QUERY strings. See: OSVDB-12184
+ /phpMyAdmin/changelog.php: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ /phpMyAdmin/changelog: Server may leak inodes via ETags, header found with file /phpMyAdmin/changelog, inode: 92462, size: 40540, mtime: Tue Dec 9 12:24:00 2008. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418
+ /phpMyAdmin/changelog: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ /test/: Directory indexing found.
+ /test/: This might be interesting.
+ /phpinfo.php: PHP is installed, and a test script which runs phpinfo() was found. This gives a lot of system information. See: CWE-552
+ /icons/: Directory indexing found.
+ /icons/README: Apache default file found. See: https://www.vntweb.co.uk/apache-restricting-access-to-iconsreadme/
+ /phpMyAdmin/: phpMyAdmin directory found.
+ /phpMyAdmin/documentation.html: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts.
+ /phpMyAdmin/README: phpMyAdmin is for managing MySQL databases, and should be protected or limited to authorized hosts. See: https://typo3.org/
+ /wp-config.php: wp-config.php file found. This file contains the credentials.
+ 8930 requests: 0 error(s) and 22 item(s) reported on remote host
+ End Time: 2026-08-10 23:41:41 (GMT+4) (112 seconds)

+ 1 host(s) tested

kali@kali:~$
```

Major Findings

- Outdated Apache 2.2.8
- Outdated PHP 5.2.4
- Missing X-Frame-Options
- Missing X-Content-Type-Options
- Apache MultiViews enabled
- HTTP TRACE method enabled
- /phpinfo.php exposed
- Directory indexing enabled for /doc/, /test/, and /icons/
- /phpMyAdmin/ exposed

Security Risks

- The outdated Apache and PHP versions may contain known vulnerabilities and increase the possibility of attacks.
- The exposed phpinfo.php can reveal detailed server and PHP configuration information.

- Exposed phpMyAdmin provides access to a database administration interface and should not be publicly accessible without proper restrictions.
- Directory indexing can allow attackers to discover files and resources.
- Missing security headers and enabled HTTP TRACE also increase the server's attack surface.

Mitigation Strategies

- Upgrade Apache and PHP to supported versions.
- Remove unnecessary backup, test, and documentation files.
- Remove or restrict access to phpinfo.php.
- Restrict phpMyAdmin to authorized users/networks.
- Disable directory listing where unnecessary.
- Disable HTTP TRACE.
- Disable MultiViews if not required.
- Configure security headers such as X-Frame-Options and X-Content-Type-Options.
- Regularly patch and scan the web server.

Conclusion

The Nikto assessment revealed weaknesses involving legacy software, exposed administration and configuration resources, directory browsing, and missing security protections. The exercise shows how initial service identification can be followed by a detailed web-server security review.

- Using Netcat (nc), investigate the host 192.168.1.40 by connecting to open ports and collecting available service banners. Analyze the information obtained and explain how attackers could use banner information during the reconnaissance phase of an attack.

Service Banner Enumeration Using Netcat

Objective

To inspect the authorized host with Netcat, identify reachable services, collect their banners, and explain how the returned information can support security reconnaissance.

Procedure

1. **Target identification:** The IP 192.168.1.40 given in the question was not reachable, so a local Metasploitable 2 lab machine was used as the authorized target. Its IP address was identified as: 192.168.56.102

2. **Connectivity test :** ping -c 4 192.168.56.102

Result :

The connectivity test received responses without packet loss, confirming communication between Kali Linux and the authorized Metasploitable host.

3. **Open-port scanning using Netcat :** nc -zv 192.168.56.102 1-1000

Port	Port name	Port	Port name
21	FTP	80	HTTP
22	SSH	111	SunRPC
23	Telnet	139	NetBIOS
25	SMTP	445	Microsoft-DS
53	DNS	512	exec

4. Banner collection:

- FTP banner was collected using : nc -v 192.168.56.102 21

Result : 220 (vsFTPd 2.3.4)

- SSH banner was collected using : nc -v 192.168.56.102 22

Result :

SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu

- HTTP was also accessed using : nc -v 192.168.56.102 80.

Result :

The response confirmed that the HTTP service was reachable on the tested port.

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=27.9 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=7.04 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=51.4 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=9.63 ms

--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3014ms
rtt min/avg/max/mdev = 7.038/23.993/51.377/17.743 ms

(kali@kali)~$ nc -zv 192.168.56.102 1-1000
192.168.56.102: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.56.102] 514 (shell) open
(UNKNOWN) [192.168.56.102] 513 (login) open
(UNKNOWN) [192.168.56.102] 512 (exec) open
(UNKNOWN) [192.168.56.102] 445 (microsoft-ds) open
(UNKNOWN) [192.168.56.102] 139 (netbios-ssn) open
(UNKNOWN) [192.168.56.102] 111 (sunrpc) open
(UNKNOWN) [192.168.56.102] 80 (http) open
(UNKNOWN) [192.168.56.102] 53 (domain) open
(UNKNOWN) [192.168.56.102] 25 (smtp) open
(UNKNOWN) [192.168.56.102] 23 (telnet) open
(UNKNOWN) [192.168.56.102] 22 (ssh) open
(UNKNOWN) [192.168.56.102] 21 (ftp) open

(kali@kali)~$ nc -v 192.168.56.102 21
192.168.56.102: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.56.102] 21 (ftp) open
220 (vsFTPd 2.3.4)
^C

(kali@kali)~$ nc -v 192.168.56.102 22
192.168.56.102: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.56.102] 22 (ssh) open
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
^C

(kali@kali)~$ nc -v 192.168.56.102 80
192.168.56.102: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.56.102] 80 (http) open
GET / HTTP/1.0

HTTP/1.1 200 OK
Date: Tue, 11 Aug 2026 04:59:44 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
```

```
Session Terminal Emulator
Use the command line
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
^C

(kali@kali)~$ nc -v 192.168.56.102 80
192.168.56.102: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.56.102] 80 (http) open
GET / HTTP/1.0

HTTP/1.1 200 OK
Date: Tue, 11 Aug 2026 04:59:44 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Connection: close
Content-Type: text/html

<html><head><title>Metasploitable2 - Linux</title></head><body>
<pre>
  _____
 /  _  _  _  \
|  _ \| | | | | |
| |_) | | | |
|  _ \| | | |
|_| \_| |_|_|
              \

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

</pre>
<ul>
<li><a href="/twiki/">TWiki</a></li>
<li><a href="/phpMyAdmin/">phpMyAdmin</a></li>
<li><a href="/mutillidae/">Mutillidae</a></li>
<li><a href="/dwa/">DWA</a></li>
<li><a href="/dav/">WebDAV</a></li>
</ul>
</body>
</html>

(kali@kali)~$ nc -v 192.168.56.102 25
192.168.56.102: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.56.102] 25 (smtp) open
220 metasploitable.localdomain ESMTP Postfix (Ubuntu)
```

Analysis

The banners exposed service identities, software products, and version details. During reconnaissance, this information can help identify technologies and compare their versions with known weaknesses, allowing an attacker to understand possible attack paths.

Conclusion

The Netcat exercise identified reachable services and collected FTP and SSH banner information. It demonstrated how service details can support reconnaissance. Administrators should maintain updated services, remove unnecessary exposure, restrict ports, and reduce unnecessary version disclosure.

4. Using Telnet, examine the server 192.168.1.50 to determine whether remote administration services are accessible. Identify any security concerns related to the discovered services and recommend more secure alternatives where appropriate.

Remote Administration Service Assessment Using Telnet

Objective :

To verify remote administration access on the authorized lab server, examine the security risks of Telnet, and identify a safer administration method.

Scanning machine: Kali Linux

Tool : Telnet

Target: Metasploitable 2

Target IP: 192.168.56.102

Procedure :

- Identified the IP address of the Metasploitable machine.
- From Kali Linux, tested the Telnet service : telnet 192.168.56.102 23
- The connection was successful and the Metasploitable login prompt was displayed.
- Tested RDP : telnet 192.168.56.102 3389
- The connection was refused, indicating that RDP was not accessible.

Commands :

- telnet 192.168.1.50 22
- telnet 192.168.1.50 23
- telnet 192.168.1.50 3389

Results :

Port	Service	Result
23	Telnet	Accessible
3389	RDP	Not Accessible

Security Concerns

- Telnet is insecure because it transmits login credentials and session data without encryption.
- An attacker monitoring the network could potentially capture sensitive information.
- Exposed remote-administration services can increase the risk of unauthorized access.

Recommendations

- Disable Telnet wherever possible.
- Use SSH (Port 22) for secure, encrypted remote administration.
- Restrict remote-administration services using firewalls and trusted networks.
- Use strong authentication and avoid exposing management services unnecessarily.

Conclusion

The assessment showed that Telnet on port 23 was accessible while RDP on port 3389 was not. Telnet is a security concern because its communication is not encrypted. SSH is therefore the recommended alternative for secure remote administration.

5. Using Netdiscover, perform host discovery on the network segment 192.168.1.1–50. Identify active hosts on the network and analyze how the discovered information could assist an attacker in planning further enumeration or exploitation activities..

Host Discovery Using Netdiscover

Objective :

To discover responding systems on the authorized lab network with Netdiscover and record their basic network information.

Tools Used

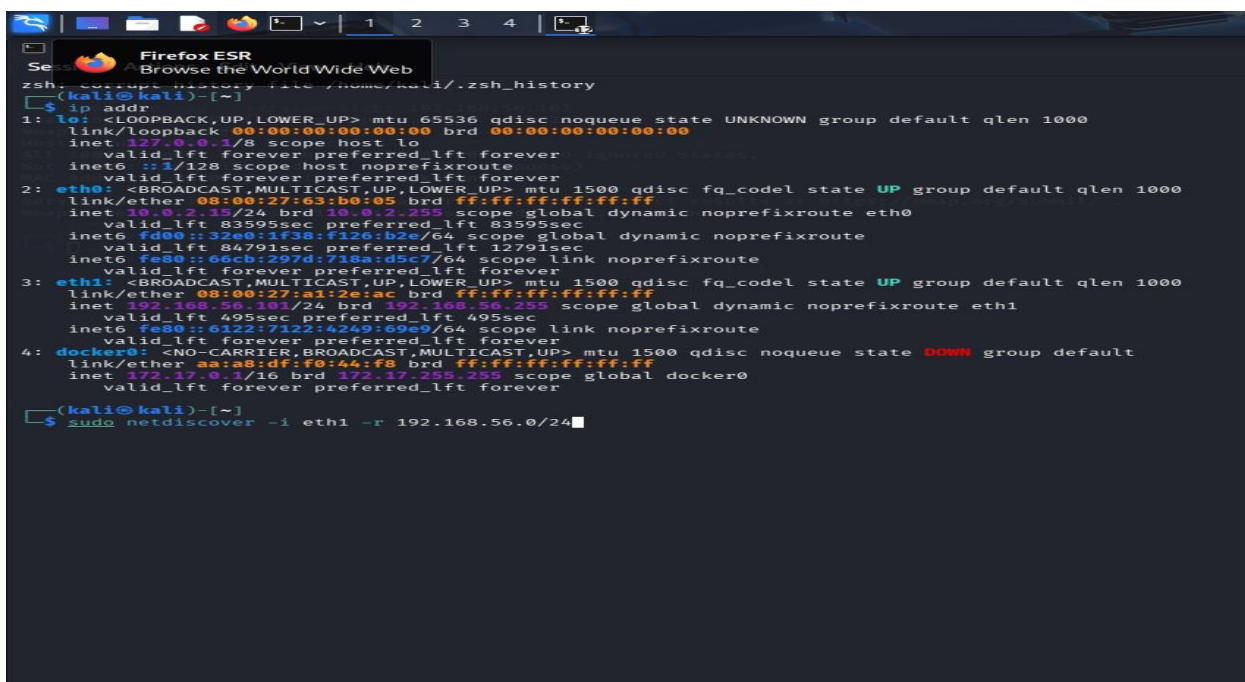
Kali Linux , Netdiscover , VirtualBox

Procedure

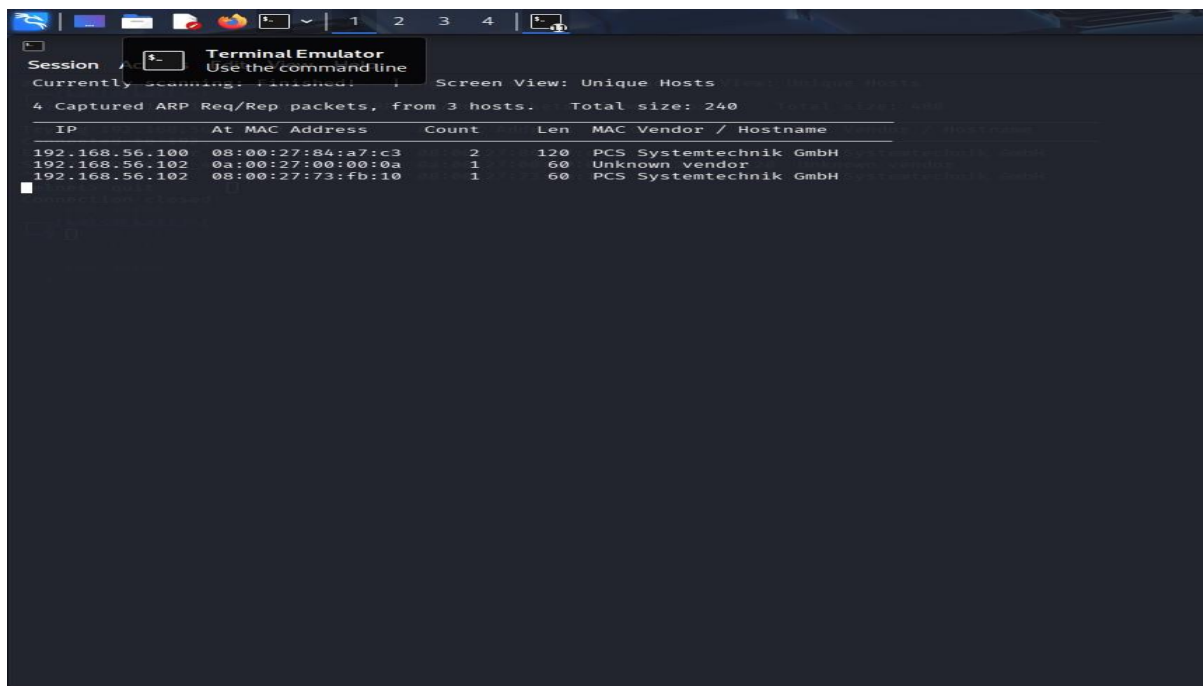
- Checked the available network interfaces using : ip addr
- Identified eth1 as the lab interface with IP address 192.168.56.101/24.
- Performed host discovery using : sudo netdiscover -i eth1 -r 192.168.56.0/24
- Waited for the scan to finish and recorded the discovered IP addresses, MAC addresses, and vendor information.

Commands :

- ip addr
- sudo netdiscover -i eth1 -r 192.168.56.0/24 | tee netdiscover_result.txt



```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 83595sec preferred_lft 83595sec
    inet6 fd00::32e8:1f38:f126:b2e/64 scope global dynamic noprefixroute
        valid_lft 84791sec preferred_lft 12791sec
    inet6 fe80::66cb:297d:718a:d5c7/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a1:2e:ac brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
        valid_lft 495sec preferred_lft 495sec
    inet6 fe80::6122:7122:4249:89e9/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether aa:a8:df:f0:44:f8 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
(kali@kali)~$ sudo netdiscover -i eth1 -r 192.168.56.0/24
```



Results :

IP Address	MAC Address	Vendor
192.168.56.100	08:00:27:84:a7:c3	PCS Systemtechnik GmbH
192.168.56.102	08:00:27:00:00:0a	Unknown Vendor
192.168.56.102	08:00:27:73:fb:10	PCS Systemtechnik GmbH

Netdiscover reported:

- 4 Captured ARP Req/Rep packets, from 3 hosts.
- The repeated 192.168.56.102 entry represents the same IP appearing with different MAC information in the virtualized lab environment.

Security Analysis

The discovered IP addresses indicate responding systems on the network, while MAC and vendor information provides additional details about their interfaces. In an authorized assessment, these details can help create a network map for later service and vulnerability checks.

Conclusion

Netdiscover provided a basic list of responding lab hosts together with IP, MAC, and vendor information. The experiment demonstrates how host discovery can create an initial network view for subsequent security assessment.

Q.No	Identification of Port Scanning Factors(25)	Analysis(25)	Linkage(20)	Application(20)	Clarity(10)	Marks
1						
2						
3						
4						
5						

Signature of Faculty

Total Avg :

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Identification of Port Scanning Factors	25%	Clearly identifies comprehensive and relevant strengths, weaknesses, opportunities, and threats	Identifies most factors with minor gaps	Limited or partially relevant identification	Incorrect or incomplete Port Scanning factors
Analysis	25%	Provides deep, insightful analysis of each Port Scanning component with strong reasoning	Provides good analysis with some depth	Basic analysis with limited reasoning	No meaningful analysis provided
Linkage	20%	Effectively links Port Scanning elements to derive strong strategic insights and implications	Shows some linkage with moderate insights	Weak linkage; limited strategic thinking	No linkage or strategic insight
Application	20%	Applies Port Scanning analysis effectively to the given case or scenario with clear relevance	Applies analysis with minor gaps	Limited application; lacks clarity	No clear application to context
Clarity	10%	Well-structured, clear, and logically presented with proper formatting	Generally clear with minor issues	Some structure but lacks clarity	Poorly organized and difficult to understand